INCH-POUND

MIL-C-71178 (AR)
10 June 1993

MILITARY SPECIFICATION

CARTRIDGE, 40mm, HIGH VELOCITY, DUMMY, M922A1 METAL PART FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the metal part for one type of cartridge designated as Cartridge, 40mm Dummy, M922A1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1310

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS

12937903 - Dummy Cartridge, 40mm, M922A1
12937904 - Linking of Dummy Cartridges, 40mm, M922A1

INSPECTION EQUIPMENT DRAWINGS

9202253 - Chamber Gage
9202254 - Limit
9202255 - Alignment

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg.) 12937903, all associated drawings, and with all requirements in applicable specifications.

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches that do not expose base material may be permitted. All required marking and stamping shall be neat and sharply defined. Total surface imperfections for nicks, dents, and scratches shall not exceed 2.5% of the surface area of the part. Any individual indentation shall not exceed .010 depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Dummy Cartridge	12937903	25

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table II) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table I below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE I. Attributes sampling inspection.

	Inspection Levels		
Lot Size	III	IV	V
2 to 8	*	*	5
9 to 15	*	13	5
16 to 25	*	13	5
26 to 50	32	13	5
51 to 90	32	13	13
91 to 150	32	13	13
151 to 280	32	32	20
281 to 500	32	32	20
501 to 1200	80	50	20
1201 to 3200	80	50	32
3201 to 10000	125	50	32
10001 to 35000	125	80	50
35001 to 150000	125	80	50
150001 to 500000	200	125	50
500001 and over	200	125	50

NOTE: Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.8). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE Dummy Cartridge	SHEET 1 OF 1	DRAWING NUMBER NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / INSPECTION METHOD REFERENCE
	Dummy Cartridge (Dwg. 12937903)		
	Examination for defects	25	3.2 Gage
	Position of ogive dot	25	3.2 4.5.2

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71178 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Dummy Cartridge 40mm, M922A1	1 OF 1	12937903
			NEXT HIGHER ASSEMBLY: 12937904

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined.			
MAJOR				
101.	Chamber gage	100%	3.2	9202253, 4.5.1 9202254, 4.5.1 9202255, 4.5.1
102.	Second largest outside diameter of cartridge - 2 places (above and below ridge)	Level III	3.2	Gage
103.	Distance from bottom of cartridge to top of taper above circular groove	Level III	3.2	Gage
104.	Spacing between ogive grooves	Level III	3.2	Gage
105.	Thickness of base rim	Level IV	3.2	Gage
106.	Ogive diameter, minimum	Level IV	3.2	Gage
107.	Distance from bottom of cartridge to top of ridge	Level IV	3.2	Gage
108.	Position of ogive dot	[1]	[1]	Visual/Manual, 4.5.2
MINOR				
201.	Poor workmanship	Level V	3.3	Visual

NOTES:

[1] One cartridge out of every 30 shall be subjected to this test. The link shall be placed on the cartridge in accordance with the procedures on drawing 12937904. The representative quantity shall be rejected if ogive dots are improperly placed (in between wrong two ogive grooves).

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4.4.3 Chamber gage. Each cartridge shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Chamber gage. The cartridge shall be placed in the chamber gage specified on the equipment list and must meet the requirement specified on the applicable drawing. This test is a non-destructive test. Parts tested may be returned to the lot.

4.5.2 Position of ogive dots. A link shall be placed on the cartridge in accordance with the procedure Notes 1 and 2 (Figures 1 and 2) on Dwg. 12937904. The ogive dot shall be visually inspected for alignment with male and female ends of the link. If the dots are between the wrong 2 ogive grooves the representative quantity shall be rejected. This test is a non-destructive test. Parts tested may be returned to the lot. (A link without a coupling may be used for this test.)

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. Preservation and packaging of the linked cartridges shall be in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level B. The cartridges shall be packed in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

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5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the metal part for Cartridge, 40mm, Dummy M922A1.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of results of contractor - conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARDEC, ATTN: SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridges are procured by AMCCOM, one (1) copy each of all ammunition data cards shall be forwarded to: Commander, ARDEC, ATTN: SMCAR-CCL-CF and SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

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6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.7 Subject term (key word) listing.

M16A2 Metallic Link
MK19 Mod 3 Grenade Machine Gun

6.8 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A533)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71178 (AR)	930610

3. DOCUMENT TITLE
CARTRIDGE, 40MM, HIGH VELOCITY, DUMMY, M922A1 METAL PART FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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